

Single Stage - Coolant Pump - Troubleshooting Guide - NGC

 haascnc.com/service/troubleshooting-and-how-to/troubleshooting/standard-coolant---troubleshooting-guide---ngc.html



Recently Updated Last updated: 10/30/2025

TG0138

Revision A - 12/2025

Download and fill out the coolant pump Inspection Report Checklist below before replacing any parts.

Introduction

The following service video shows how to troubleshoot a standard flood coolant pump

NOTE: This video is for reference only and does not replace the written procedure.

This troubleshooting guide applies to both 1-Phase and 3-Phase pumps.

In this image, the following components are shown.

1. Coolant Pump Motor
2. Coolant Outlet
3. Pump Body
4. Impeller Cap
5. Impeller

NOTE: For troubleshooting information on the coolant float sensor, refer to the [Coolant Level Float Sensor - Troubleshooting Guide](#) for more information.

NOTE: Coolant Pumps with **PN: 30-14261** (1HP, 3-Phase without VFD) will follow the same installation process for non-VFD applications across all machines, with the exception of VF's VP-5, VM's, ST/DS, and UMC's.

ID0013 Coolant Pump Inspection Report Checklist requires the pump's serial number to be completed.

The serial number can be found on the side of the pump [1] in the box labeled serial number [2]

Symptom Table

Symptom	Possible Cause	Corrective Action
Machine with Variable Frequency Drive - The coolant takes too long to flow out the nozzles or Pcool.	The variable frequency drive has an incorrect acceleration parameter value.	Refer to the Coolant VFD - Troubleshooting Guide .
Noisy pump or cavitation	Poor Coolant flow through enclosure	Remove chip buildup.
	Low Coolant Level	Fill coolant tank.
	Impeller contamination, or impeller damage	Inspect impeller for damage or debris. Clean impeller or order kit 93-2609 to replace impeller. Follow Impeller Replacement Procedure to replace.
	Motor bearing failure.	Rotate the shaft by hand

Low coolant flow.	Poor Coolant flow through enclosure	Remove chip buildup.
	Low coolant level in coolant tank	Fill coolant tank.
	Impeller contamination, or impeller damage	Refer to the Coolant Pump Impeller Damage/Contamination section below to check the impeller in the pump.
	Filter or pump intake blockage.	Clean filter, check pump intake for blockage.
	Pump intake or coolant output hose blockage	Remove coolant check-valve and coolant output hose blockage.
	Coolant nozzle blockage	Remove blockage from coolant nozzles
The coolant pump is not building pressure.	The motor shaft may be slipping due to the coupling not being fastened all the way.	Thoroughly inspect the pump assembly and verify the motor shaft coupling is fastened down all the way.

No coolant flow and coolant circuit breaker did not trip.	Setting 32 COOLANT OVERRIDE is set to IGNORE.	Set Setting 32 COOLANT OVERRIDE to NORMAL.
	Coolant pump motor does not receive voltage.	See the No Voltage at the Pump Motor section below.
	Incorrect power phasing - pump motor is running backward.	Correct phasing
	Impeller contamination or damage	Refer to the Coolant Pump Impeller Damage/Contamination section below to check the impeller in the pump.
No coolant flow and the coolant circuit breaker tripped.	Coolant pump motor short to ground.	Measure coolant pump motor resistance to ground.
(Single-phase machines only), Alarm 552 TRIPPED CIRCUIT BREAKER occurs when the coolant pump gets commanded to turn on. Note: The circuit breaker did not tripped.	The 240 VAC input cable connected to the PSUP at P11 is not correct.	Verify the cable pinout for the cable that connects from the main circuit breaker to the PSUP at P11. Refer to Incorrectly pinned power cable (Single-Phase) section.

Alarm 552 TRIPPED CIRCUIT BREAKER	Refer to: Next Generation Control - Power Distribution PCB and PFDM - Troubleshooting Guide.	
Coolant leaking from machine enclosure onto floor.	This applies to ST-20 reboot enclosures only. The grommet that seals the Sub-Spindle Hydraulic Union drain line fails to seal due to the hose pulling on the grommet.	Install the ST-20 Sub-Spindle Hose Routing Retrofit Kit Part Number: 93-3127. This reroutes the drain hose and clamps the hose limiting the movement and ability to pull on the grommet. Follow the Installation Guide to retrofit your machine.
Alarm 9823 Coolant VFD Fault Alarm 9824 Shower Coolant VFD Fault	There is a problem with the Coolant / Shower VFD.	Refer to the Coolant VFD - Troubleshooting Guide.
Coolant pump with an Aux filter is splashing on the floor when pump is commanded off	When pump is commanded to stop coolant left over in the line returns through the pump and splashes into the tank.	Install a check valve on the output side of the aux filter to stop the coolant from returning and splashing. Please refer to <u>Aux Filter Check Valve</u> section for installation. Please contact Haas Service if a check valve is installed to the output of the aux filter.

Motor hums when commanded to turn on	Note: For pumps dated DEC-2019 to FEB-2020 Coolant pump start capacitor failed.	Refer to the Coolant Pump - Start Capacitor - Replacement procedure.
Turns slowly, drawing higher than normal current and eventually trips the breaker		
Turns backwards		
Alarm 9823 COOLANT VFD FAULT	The flood coolant pump has seized up.	Refer to the Coolant Pump Impeller Damage/Contamination section below to check the impeller in the pump.
Alarm 552 TRIPPED CIRCUIT BREAKER and The CB3 on the PSUP PCB has tripped on machines without a VFD.		
The 5/8" coolant hose leaks at the fittings or the hose pops off the fittings.	The 5/8" Hose Clamp[1] PN: 59-1524 installed on the coolant hose is oversized and does not clamp the hose correctly.	Replace the oversized hose clamp with PN: 59-2107 HOSE CLAMP EAR 20.1 - 23.3 MM. Use a Crimping Tool to secure the new hose clamp.

Poor coolant flow through enclosure

Remove all chips from the the auger trough [1], and the coolant return drain [2].

Remove chips from the coolant tank. Excessive chips in the coolant tank reduce the amount of coolant the tank can store. Watch the video [Machine Tool Coolant - Cleaning Your Tank - Video](#).

Fill the coolant tank. Watch the video [Standard Coolant – Simplified - Filling your Tank ... and More](#).

Make sure the coolant level sensor operates correctly.

Coolant Pump Impeller Damage/Contamination

Remove the impeller cap [1] from the bottom of the pump. Remove the screw that connects the impeller [2] to the shaft. Steel impellers have left-hand threads [3].

Carefully remove and clean the impeller [2]. Check for wear, damage, or cracking of the impeller [2].

If the impeller is damaged or cracked, order a replacement impeller, service part **93-2609**. Follow [Impeller Replacement Procedure](#) to replace.

Rotate the coolant pump shaft by hand. If the shaft does not turn freely, the motor bearings are damaged.

Install the impeller [2]. Rotate the coolant pump shaft by hand, and make sure the impeller does not rub on the pump body [4]. If the impeller rubs against the pump body, install shims between the impeller [2] and the coolant pump shaft. Install the impeller cap [1]. If the impeller [2] rubs against the impeller cap [1], remove one shim at a time until it does not rub.

There are several filters that keep chips out of the coolant pump. Refer to Section 3 for filter maintenance information.

Aux Filter Check Valve Install



Place the check valve [1] on the outlet side of the auxiliary filter as shown.

NOTE: Please contact Haas Service if this is done.

Filter or pump blockage

Clean or replace the filter element. Check the area around the coolant pump intake for chips.

1. Auxiliary Filter - The bag filters in Auxiliary Filter tanks need to be cleaned or replaced. The Auxiliary Filter is an option. Your machine might not have this option.
2. Standard Filter - Standard Filters need to be cleaned.

Disconnect both ends of the coolant hoses. Remove any kinks. In the opposite direction of the coolant flow, blow air through the hoses to remove any chips. If you find chips in the hoses, check the coolant nozzles and washdown hoses for chip buildup. Connect the coolant hoses.

Remove chips from the basket filter in the coolant tank. For maintenance instructions, refer to the [Coolant Tank - Maintenance](#) procedure.

Check valve blockage (Coolant filter and Haas Chip Separator)

Remove the coolant check-valve. Check it for blockage [1]. Make sure the trap door opens and closes freely. If the trap door does not close, coolant flows back into the tank. This delays the flow of coolant by a few seconds.

Disconnect both ends of the coolant hoses. Remove any kinks. In the opposite direction of the coolant flow, blow air through the hoses to remove any chips. If you find chips in the hoses, check the coolant nozzles and washdown hoses for chip buildup. Connect the coolant hoses.

Coolant nozzle blockage

Remove and clean the coolant nozzles.

Cycle the pump power to flush the coolant lines.

Make sure the coolant tank is free of chips and contamination.

Defective pump

1. Disconnect the pump cable from the machine.
2. With a ohm-meter measure each leg tab to the ground tab.
Note: The ground tab has a bent. The example shown is a 3-phase pump with 4 tabs (L1, L2, L3 + Ground), single phase pumps have 3 tabs (L1, L2 + Ground).
 - If all the legs test open (O.L) to the ground tab then the pump and cable do not have a short circuit, see the next section.
 - If the motor cable leads [3] indicate a short, remove the motor cable cover and disconnect the cable from the motor.
3. Measure the motor cable leads [5] to ground [6] terminal screw or chassis. If any lead tests short, the pump is at fault. If none of the leads [5] test short, the cable is at fault.
4. If all the leads test OK, check the coolant pump for binding that could cause an over amperage condition.

No voltage at pump motor

1. First inspect the pump motor for a short circuit see section **Defective Pump**.
2. Make sure the coolant pump cable is connected to the machine.
Important: Do not disconnect the coolant pump from the machine, doing this test without a load will give you false readings.
3. Remove the pump motor cover and locate the white, black, red wires for 3-phase motor or white and black for a single-phase.
4. Verify that coolant circuit breaker is set to on and **setting 32 COOLANT OVERRIDE** is set to NORMAL.
5. Press the **[COOLANT]** button to turn on the coolant pump.
6. Measure the voltage at the motor from lead to lead [1]. The reading must be 240 VAC. The motor shown is a 3-phase motor.
 - If you read 240 VAC, the coolant pump motor is damaged.
 - If you do not read 240 VAC, check at the I/O PCB coolant output. If there is still no voltage check the 240 VAC source from the PSUP PCB to I/O PCB.

Incorrect phasing

The power supply PCB has a phase detect with neon indicators on the top center portion of the board. Make sure that the electrical power is phased correctly:

- **Green Light:** The incoming power is phased correctly.
- **Orange Light:** The incoming power is incorrectly phased.
- **Both Lights:** A phase is missing (there is a loose cable in the system).

If the electrical power is phased incorrectly:

Set the main circuit breaker to the OFF position.

Lock the main circuit breaker. Use an approved lock with an approved safety tag.

Swap the #74 and #75 incoming power cables at the main transformer.

Incorrectly pinned power cable (Single-Phase)

Correct the pinnout on cable P/N 33-0197C, PSUP P11 (NGC).

The cable should match the drawing shown.

Coolant Hose Clamp Failure

Problem: The 5/8" coolant hose leaks at the fittings or the hose pops off the fittings.

Cause: The 5/8" Hose Clamp[1] PN: **59-1524** installed on the coolant hose is oversized and does not clamp the hose correctly.

Solution: Replace the oversized hose clamp with PN: **59-2107** HOSE CLAMP EAR 20.1 - 23.3 MM. Use a [Crimping Tool](#) to secure the new hose clamp.

Electrical Diagrams

Use this electrical diagram for coolant pumps that are being driven by the I/O PCB.

